

Revisiting Aerosol Budgets under Seasonal Constraints

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Abstract

We examine the behaviour of runoff models in urban heat islands. The reduced model retains most of the explanatory power at a fraction of the computational cost. We release the processing scripts and intermediate artefacts to support independent verification. A modest but repeatable bias appears in the lowest-magnitude bin and warrants further study.

1. Introduction

The reduced model retains most of the explanatory power at a fraction of the computational cost. Neither of the alternative formulations improved on the simple additive model in our setting. We observe a pronounced inflection once the load exceeds roughly sixty percent of nominal capacity. Our baseline reproduces the behaviour described in earlier work to within the stated tolerance.

Residual error grows approximately linearly with the number of coupled parameters. We release the processing scripts and intermediate artefacts to support independent verification. Our baseline reproduces the behaviour described in earlier work to within the stated tolerance. A modest but repeatable bias appears in the lowest-magnitude bin and warrants further study. Discrepancies at the boundary are consistent with the simplifying assumptions noted above. The measurements reported here were collected over twelve independent trials under controlled conditions. We observe a pronounced inflection once the load exceeds roughly sixty percent of nominal capacity. Replicate agreement was high, suggesting the protocol is robust to small handling differences.

2. Related Work

Sensitivity to the smoothing window proved modest across the range we examined. Taken together, these findings argue for treating the two regimes separately in future analyses. Our baseline reproduces the behaviour described in earlier work to within the stated tolerance. Neither of the alternative formulations improved on the simple additive model in our setting.

The measurements reported here were collected over twelve independent trials under controlled conditions. Our baseline reproduces the behaviour described in earlier work to within the stated tolerance. Replicate agreement was high, suggesting the protocol is robust to small handling differences. Calibration drift over the trial period remained below the threshold set out in the protocol. Longer observation windows narrow the

confidence interval but do not shift the central estimate. The reduced model retains most of the explanatory power at a fraction of the computational cost. Taken together, these findings argue for treating the two regimes separately in future analyses. Discrepancies at the boundary are consistent with the simplifying assumptions noted above. Neither of the alternative formulations improved on the simple additive model in our setting.

3. Materials and Methods

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4. Experimental Setup

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